

# Design Patterns of Technology-based Therapeutic Activities for Children with Language Impairments: A Psycholinguistic-Driven Approach

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## ABSTRACT

In the last years, researchers and therapists have pinpointed a number of critical aspects in current speech-language interventions. Several studies have explored the use of technology to overcome these barriers and to support speech-therapy in children with language impairments (e.g. DLD and ASD). In this paper, we propose a conceptual framework for designing linguistic activities (for assessment and training), based on advances in psycholinguistics. Moving from this theoretical framework, we identified a development process - from the UX design to coding of activities - which is based on a novel set of Design Patterns at multiple layers of abstraction. We then put this framework into practice by implementing these patterns into two technological solutions - tablet and robots - and performing an empirical study to evaluate their benefits. Our results, although still preliminary since they assessed only a tablet experimental condition, are promising for extending the identified Design Patterns to other technological solutions.

## CCS CONCEPTS

• **Human-centered computing** → **HCI design and evaluation methods**; *Empirical studies in HCI*.

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*CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan*

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ACM ISBN 978-1-4503-8095-9/21/05...\$15.00

<https://doi.org/10.1145/3411763.3451775>

## KEYWORDS

platform-agnostic, speech-language therapy, children with DLD and ASD, adoptability, multi-modality, multi-channel, social assistive robots

### ACM Reference Format:

Micol Spitale, Silvia Silleresi, Giulia Leonardi, Fabrizio Arosio, Beatrice Giustolisi, Maria Teresa Guasti, and Franca Garzotto. 2021. Design Patterns of Technology-based Therapeutic Activities for Children with Language Impairments: A Psycholinguistic-Driven Approach. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3411763.3451775>

## 1 INTRODUCTION AND BACKGROUND

Speech and language impairments are among the most prevalent childhood disabilities worldwide [40]. In 2018, epidemiological studies have reported that 10% of pre-schoolers and 5-6% of 5-18 years old children/adolescents have shortcomings in language abilities [23]. Difficulties in language development, especially on structural aspects of language (morphosyntax and phonology) have been observed in a broad range of neurodevelopmental disorders. For some of these conditions structural language impairments represent the primary domain of impairment, e.g. Developmental Language Disorder (DLD); for other conditions, e.g., Autism Spectrum Disorders (ASD) it is diagnosed as a possible secondary accompanying deficit [38] [39] [27]<sup>1</sup>.

<sup>1</sup>DLD, whose prevalence rate is approximately 7.5% [37], is defined as a neurodevelopmental disorder that involves specific difficulties in mastering structural aspects of language (morphosyntax and phonology), independently from any kind of intellectual disability, sensory impairment or neurological dysfunction [33]. ASD, which has an estimated prevalence of around 1%; [7], is diagnosed when a child shows persistent deficits in social communication and interaction, as well as restricted and repetitive behaviors or interests [6]. Children who meet criteria for ASD may have accompanying language impairments, which affects specific aspects of structural language

In recent years, researchers and therapists have pinpointed a number of critical aspects in current speech-language therapy, in particular with respect to children with language disorders. First, some scholars suggested that the assessment tools currently used in clinical and research practice may not be adequate for actually evaluating language skills in children with structural language impairments, as they often lack attractiveness and engagement, and are perceived as stressful and frustrating, with the result of enhancing children's weaknesses rather than their points of strengths [62], [53]. Second, the American Speech-Language-Hearing Association (ASHA) [5] pinpointed that current therapeutic practices would need a more holistic perspective in which the goal of language training is to stimulate linguistic development alongside with extra-linguistic skills, such as cognitive aspects of communication (e.g., attention, memory, sequencing, problem-solving, executive functioning) and social aspects of communication (including behavior). Third, the World Health Organization has identified significant barriers of access to treatment, in terms of costs, long-term involvement, and difficulty to extend interventions beyond the clinical context (e.g., home and school) [23].

A number of studies has explored the introduction of interactive technologies in speech-language therapy as a means to overcome some of the above limitations. Interactive technologies have the potential to attract and engage children with speech and language impairments more than conventional intervention tools [12], [17] and seem to be effective to acquire language skills faster and retain language skills longer [21].

The most promising approaches involve regular devices (computers, tablets or smartphones, hereinafter referred to as "computers") and robots. In fact, many studies provided evidence not only of the effectiveness of computer-based applications for training linguistic abilities of people (children and adults) with language disorders (e.g., [31], [36] [55], [1], [13], [22], [51]), but also their potentially low barriers for adoption, because caregivers (therapists and families) are usually familiar with them, and these devices are relatively low cost [35]. In terms of attractiveness and engagement, the introduction of robots in speech-language therapy (i.e., Socially Assistive Robots [24]) for children with language impairments seems to have even higher benefits than computers, due to the embodiment characteristics of robots [16]. In this vein, several studies provided evidence of the efficacy of robots not only for improving speech-language skills but also for supporting extra-linguistic skills (e.g., executive functions, communication skills, joint attention, turn-taking, and eye-contact [48], [49], [10], [50], [11], [14]).

Our work explores the use of the two above mentioned classes of technologies - computers and robots - for speech therapy, focusing on children with DLD and ASD. We have designed and implemented a set of interactive therapeutic activities for this target group that are characterized by a number of unique features:

- Activities are both inspired by speech-therapy practice and grounded on advances of psycholinguistics research (i.e., their design is informed on the identification of common phenotypical markers of language impairments in children with DLD and ASD [58]).

- Activities support both assessment and training of language abilities and they are integrated in a methodologically coherent technological environment.
- Activities have been defined as patterns which are platform-agnostic and multi-channel (i.e., they can be performed using any kind of technological solution)
- Activities have been implemented as one type of device only, or combination of computer and robot.

The whole development process - from the UX design of activities to coding - is based on a novel set of Design Patterns at multiple layers of abstraction. By the very nature of a pattern-based methodology, this approach facilitates design reuse and extensibility at both UX and implementation level [9, 42, 52, 59]. Our innovative approach addressed the above-mentioned clinical interventions barriers and meet linguistic requirements of children with structural language impairment. The primary contribution is the design and development process of the robot and computer-based activities for training and assessing linguistic capabilities of children with structural language impairments, e.g., DLD and ASD (language impairments from now on).

The rest of the paper presents the design of our activities, an empirical study - currently ongoing - devoted to evaluate the effectiveness of our linguistic activities for children with DLD and ASD, and discuss the results achieved so far.

## 2 DESIGN PROCESS

The co-design process involved a multidisciplinary team of computer engineers, interface designers, psycholinguistic researchers, and speech-language therapists specialized in the treatment of children with DLD and ASD. The process unfolded along two main steps: the definitions of i) the psycholinguistic methodological framework and ii) Activity Patterns.

### 2.1 Psycholinguistic methodological framework

**2.1.1 Theoretical Background.** Some studies in psycholinguistics have investigated the existence of common or unique linguistic profiles among subjects with DLD and ASD, and whether they can be associated to shortcomings on structural aspects of language [27], [8], [34]. In this regard, despite DLD and ASD are mutually exclusive diagnoses [6] and the precise etiology of linguistic difficulties seem to differ among these two groups, several studies have found in both these conditions common phenotypical traits of impairments on specific linguistic constructions or tasks, called *markers*, that are good indicators of underlying structural language difficulties [15]. In particular, regarding morphosyntactic abilities, impairments have been attested on: (1) passive clauses (e.g., "the princess is covered by the soldier") [19], [57], [26], [45], [47]); (2) clitic pronouns (e.g., "In this story there are a Queen and a King. Look! Mary is touching her") [18], [44], [41], [28], [2]; (3) relative clauses among which we differentiate: Passive Object Relatives (e.g., "this is the horse that is pulled by the cows"), Object Relatives (e.g., "this is the horse that the cows are pulling") and Subject Relatives (e.g., "this is the horse that is pulling the cows") [46], [56], [20], [61], [4]; (4) Wh-questions among which we differentiate: Object-Wh-questions (e.g., "which is the dog that the turtles are washing?")

development. It is estimated that 50%-70% of verbal children with ASD show deficit in structural aspects of language [60]

and Subject-Wh-questions (e.g., “which is the dog that is washing the turtles?”) [43], [25].

**2.1.2 Framework-driven speech therapy activities.** To evaluate (assess) the ability of comprehending or producing these morphosyntactic constructions, psycholinguistic research has defined several tasks targeting markers of language impairments. In these tasks, which are usually paper-based, children are presented with a series of contents (spoken sentences and/or printed images) and they are asked to comprehend/produce a specific morphosyntactic structure. For example, in a comprehension task a sentence is told and two or more pictures are shown; children have to find among the pictures the one that represents the sentence they heard. Crucially, for assessment activities, each combination of sentences and/or images is presented as an isolated task and it is independent from the linguistic context. Moreover, due to the evaluative nature of the activity, no negative or positive feedback is given to children, due to the evaluative nature of the activity. Since the use of these theory-driven activities has been proved very effective for assessment purposes, the psycholinguists of the team created a new set of activities, with the specific purpose of training the morphosyntactic constructions targeting markers of language impairments, extending the psycholinguistic framework to this novel application domain. For training activities, we applied the “storytelling” format because it is widely demonstrated that the use of a linguistic context (as in storytelling) influences and boosts children’s language skills and syntactic comprehension [32]. The task is conceptually very similar to the activity proposed in assessment, except for the storytelling context and the feedback provided. In fact in training activities, children are presented with several combinations of contents wrapped up in a story, and are provided with negative and positive feedback to train effectively their skills. Moving from this theoretical framework we proceeded to the design phase of assessment and training activities. The set of linguistic activities was composed by two subsets: the first one was devoted to assessment purposes; the second subset to training purposes. Both subsets included activities that targeted comprehension (i.e., children’s capability to understand what they hear) and production skills (i.e., children’s capability to express themselves by speech). All the tasks were in Italian language. Table 1 reports the set of linguistic activities designed within the theoretical framework. The design of assessment activities resulted in 12 tasks (8 for comprehension and 4 for production). These activities were inspired from existing tasks already present in psycholinguistic framework. With the exception of the passive clauses task, which was specifically developed for the current research project, we adapted the relative clauses task from [29], the clitic pronouns task from [30] and the Wh-questions task from [3] and the Sentence Repetition task from [54].

Analogously, we designed the training activities which resulted in 12 tasks (8 for comprehension and 4 for production). While all the contents (sentences and images) of the comprehension activities were created from scratch, for training production skills, we re-adapted the Sequential Order Subtest of [63] by using four sets of four pictures. The four stories were created by the psycholinguists of the team, who designed them by employing the exact same narrative structure, which was composed by nine main elements (e.g. who are the main characters in the story, which is the main event,

what is the main goal of the characters, what are the emotional states of the characters) and the same targeted morphosyntactic structures.

## 2.2 Activity Patterns

The design and development corresponding to the activities defined in the methodological framework is based on a set of Design Patterns, collectively referred to as *Activity Patterns*. In many fields such as software engineering, web engineering, and Human-Computer Interaction, Design Patterns are general, reusable solutions to commonly occurring problems within a given domain, and provide blueprints or templates that can be customized in a specific context. Activity Patterns are defined at different levels of abstraction, resulting into three categories of patterns: *Abstract Task Patterns*, *Interaction Patterns*, and *Architectural Patterns* (see Figure 1). Lack of space prevents us to use the reference templates that the Design Patterns literature provide for pattern description. In the rest of this section Activity Patterns are reported informally, through a short description and a visual representation (see Figure1).

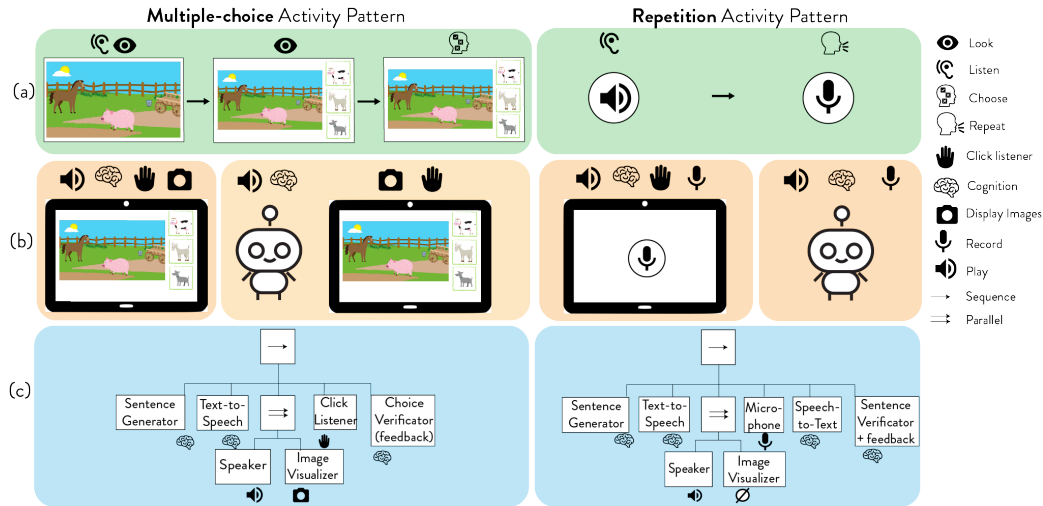
**2.2.1 Abstract Task Patterns.** We depicted these patterns in the top layer of Figure 1 (a). Abstract Task Patterns are fully technology-independent, i.e., they do not depend on the devices used and are the result of an abstraction process on the tasks specifications defined in the Section 2.1 (see Table 1). The definition of Abstract Task Patterns only refers to the types of stimuli the child receives, the reasoning and linguistic process involved, and the types of actions they have to perform. Conceptualizing these tasks in terms of the above elements, two main recurring situations emerge, both in assessment and in training, leading to two Abstract Task Patterns:

- **AT1-Multiple-choice.** This pattern is visually depicted in Figure 1 - layer (a) - left side; it conceptualizes a situation that involves user’s sight and hearing, and requires language comprehension and decision-making among multiple alternatives provided. The child is offered a sequence of audio or visual stimuli (voice, image, image+text) and has to select one “answer” among the different alternatives that are presented. This pattern is exploited for comprehension activities in both assessment and training, such as Picture Matching and Storytelling Picture-Matching. For example in the Storytelling Picture-Matching (see Table 1), the user sees a picture, depicting a scene of the story, while they hear the corresponding sentence. Then they see three pictures corresponding to three options, and has to decide which is the correct one that corresponds to the sentence just heard.
- **AT2-Repetition.** This pattern is visually depicted in Figure 1 - layer (a) - right side; it involves user hearing, reasoning, and speaking (reproducing verbally what is heard, either a sentence or a story). This pattern is used for production activities in assessment and training, such as Sentence Repetition and Retelling tasks in Table 1.

**2.2.2 Interaction Design Patterns.** Interaction Design Patterns can be regarded as the operationalization in technological contexts of what is described in Abstract Task Patterns. The description of Interaction Design Patterns include the device(s) (tablet, robot, or their combination) used for computation and stimuli generation

**Table 1: Framework-driven psycholinguistic activities**

| Goal       | Targeted Linguistic Skills | Type of Linguistic Tasks         | N of tasks and targeted morphosyntactic structures                                                            | Task Execution                                                                                                                                                                                                                        |
|------------|----------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Assessment | Comprehension              | Picture-Matching                 | - 2 for Clitic Pronouns<br>- 2 for Passive Clause<br>- 2 for Relative Clauses<br>- 2 for Wh- questions        | Children listen to a sentence while at the same time are presented with a set of pictures (2 or more). Then, they have to choose the figure that represents what they heard                                                           |
|            | Production                 | Sentence Repetition              | - 2 for Clitic Pronouns and Passive Clauses<br>- 2 for Relative Clauses and Wh- questions                     | Children listen to a sentence and they have to repeat the sentence verbatim (identical to the target sentence)                                                                                                                        |
| Training   | Comprehension              | Storytelling<br>Picture-Matching | - 2 for Clitic Pronouns and Passive Clauses<br>- 2 for Relative Clauses<br>- 4-levels story for Wh- questions | Children listen to a sentence and see a main scene. At the same time they are presented with three different figures. Children's task is to select the relevant figure. If they select the correct feature the story continues.       |
|            | Production                 | Retelling of the story           | - 2 for Clitic Pronouns and Passive Clauses<br>- 2 for Relative Clauses and Wh- questions                     | Children see four images in sequence while listening to a story. Then, they are asked to retell the story as close to the original as possible, using the pictures as prompts, including all the nine elements and correct structures |

**Figure 1: Activity Patterns. Layer (a): Abstract Task Patterns; Layer (b): Interaction Design Patterns; Layer (c): Architectural Patterns.**

and the user's interaction with such devices, as shown in Figure 1-layer (b). The use of different devices, or of a combination of devices, implies different interaction paradigms: touch-based GUI interaction (for the tablet or smart-phone), speech-based interaction (for the robot, communicating with it by voice), and hybrid interaction (for the combination of tablet + robot, enabling the user to interact via touch and speech). Taking different interaction paradigms into account lead to different Interaction Design Patterns related to the same Abstract Task Pattern.

- **IP1-Multiple-choice/Mobile Device.** This Interaction Design Pattern (See Figure 1 - layer (b) - left column - left side) shows that the user sees an image displayed on the screen of the device (tablet or smart-phone) while they hear something (a sentence or a story) played by the device itself, then they are presented with a set of visual options, and has to touch on the one they perceive as most appropriate for the current context. The logic of the activity (e.g., providing or not a positive feedback and moving forward in case of

correct answer, or providing or not a negative feedback and repeating the same scene in case of wrong selection) stays in the device.

- **IP2-Multiple-choice/Robot+Mobile Device.** This Interaction Design Pattern is depicted in Figure 1 - layer (b) - left column - right side. Like the situation described in IP1, in IP2 the mobile device is the GUI interface, and the locus of visual stimuli display and of touch interaction for choice selection. Still, the logic of the activity and the sound playing are managed by the robot. This Interaction Design Pattern extends the interactive experience described in IP1 by taking advantage of the expressiveness power of the robot, i.e., its capability for rendering facial expression, eye gaze, or body gestures.
- **IP3-Repetition/Mobile.** This Interaction Design Pattern is depicted in Figure 1 - layer (b) - right column - left side. It is appropriate for both Sentence Repetition and Story Retelling tasks, where the user has to repeat what they heard and the system should check the correctness of the repeated sentence. The interaction is speech and touch based. In Retelling Tasks, the user looks at the images while listening to a story. The user listens to the sound played by the mobile device (e.g., tablet), then they have to click on the screen for starting the recording of his/her voice repeating the story. The mobile device manages the interaction logic, checking the correctness of the user's speech and taking further actions according to the results (e.g., providing positive/negative feedback, asking questions to clarify the content of the story).
- **IP4-Repetition/Robot.** This Interaction Design Pattern, depicted in Figure 1 - layer (b) - right column - right side, supports exclusively speech-based interaction with the robot while performing Sentence Repetition tasks. The robot utters a sentence, and asks the child to repeat the sentence they heard. The robot stops listening to the user as soon as they stop talking, and then it analyzes the sentence just told, and moves on in the next step according to the logic of the ongoing activity.

**2.2.3 Architectural Patterns.** Architectural Patterns are the lowest level in the set of our Activity Patterns (see Figure 1 (c)). They provide general solutions to implement the situations described in the Abstract Task Patterns and provide the core elements of a modular, platform-agnostic architecture. Each Architectural Pattern step (e.g., Sentence Generator, Text-to-Speech) corresponds to a system module to be triggered when required by the activity. The different modules are dedicated to single purpose, and they are also easily composable to bring together various different capabilities to accomplish complex goals. We describe each Architectural Pattern as follows:

- **AP1-Multiple-choice.** This Architectural Pattern, which is depicted in Figure 1 - layer (c) - left column, contains the system modules. The sequence start with the Sentence Generator module which selects a sentence and then the Text-to-Speech synthesizes it. Then, the system displays an image (Image Visualizer), as it tells the first sentence of the story (i.e., play a sound, Speaker). After that, the Event Listener waits until the user chooses an answer among a series of

alternatives. In case of training activity, if the user selects the image correctly, the Choice Vericator rewards the user with a positive feedback and the story moves forward, otherwise the child is asked to repeat again the same task. While for assessment, the system moves on independently from user choice.

- **AP2-Repetition.** This Architectural Pattern is depicted in Figure 1 - layer (c) - right-side. Again, the Sentence Generator picks the sentence to be synthesized by Text-to-Speech, and the Speaker plays the audio generated. The Image Visualizer module is exploited only in activities which require a visual support. In case of Sentence Repetition, no image is displayed (Image Visualizer is empty), while in the Retelling activity, the system display an image corresponding to the story scene. The Microphone senses what the user says, and Speech-to-Text module processed the audio data, transcribing the user speech. Then, the Sentence Vericator checks if the user repeats correctly the sentence (or the story), and it provides a feedback only in the case of training (i.e., Retelling).

### 3 EMPIRICAL STUDY

We are performing an empirical study to evaluate the benefits of the activities, comparing them in different experimental conditions to: i) test the representation of these activities through a computational medium vs non computational medium, ii) compare the children performance among the computationally mediated interventions (i.e., tablet, tablet+robot, and robot). The hypothesis of our research protocol was twofold: i) children would improve their morphosyntactic scores in comprehension more via computationally mediated intervention than via non-computationally ones; ii) children would achieve better results using the robot+tablet intervention with respect to the tablet- or robot-intervention. The study is organized in 3 controlled sub-studies. Each sub-study focuses on one experimental condition (i.e., tablet, tablet + robot, or robot) and a different set of participants. For all sub-studies, we recruited a target and a control group of participants, involving four local therapeutic centres. The target groups are assessed and trained via computationally mediated intervention while the control groups via non-computational medium. The empirical work is currently ongoing. The results achieved so far concern the first sub-study (tablet only) on the target group only. Due to the pandemic, we were not able to conduct the in-person study with non-computational medium. The timeline of the research protocol (8 sessions) was the following: Assessment (Session 1); Training (Sessions 2 to 7) and Re-assessment (Session 8).

For this sub-study, we have recruited 30 children (22 with DLD and 8 with ASD). The diagnosis of developmental language impairment or autism was confirmed by the DSM-5 criteria. The participants inclusion criteria were: i) monolingual Italian-speaking children, ii) a minimum mean length of utterance (MLU) of 2.5, iii) children morphosyntactic deficit of at least  $-1.5 SD$  on morphosyntactic standardized tests, iv) children age 4-12 y.o. Due to the current pandemic situation, so far only 12 of the 30 children could start the study and complete it. The 12 children (10 with DLD and 2 with ASD) were assessed and trained on comprehension activities (Picture matching task and Picture-matching task in a Storytelling).

We chose to separate children into two groups, on the basis of their chronological age and their developmental language level (at age 8 children should have acquired passives and clitics; at age 12 relative clauses and wh-questions). Group 1 included 6 children (Mean age = 6.1 - 5 children with DLD and 1 child with ASD) trained on clitic pronouns and passive sentences; Group 2 included 6 children (Mean age = 8.6 - 5 children with DLD and 1 child with ASD) assessed and trained on relative clauses and wh-questions. Informed parental consent was obtained for all participants included in the study. The study received the approval from the Ethical Committee of the University.

### 3.1 Preliminary Results

From activity logs we gathered quantitative data concerning performance on the linguistic activities: we collected the average number of correct responses for each activity performed on the 8 sessions by each group. In order to see whether the activities were effective, we looked at the performance for each group on each targeted structure, focusing on the differences between assessment and re-assessment phases. Group 1 showed a significant improvement of performance on clitic pronouns ( $U(5) = 34.5, p = .035$ ) and on passive clauses ( $U(5) = 1, p < .001$ ). Group 2, on the other hand, showed a stable trend both on Relative Clauses (Passive Object Relatives:  $U(5) = 38.9, p = .054$ , Object Relatives:  $U(5) = 41, p = .228$ , and Subject Relatives:  $U(5) = 41, p = .090$ ) and on Wh-questions (Object Wh-questions:  $U(5) = 48.5, p = .456$ , and Subject Wh-questions:  $U(5) = 58, p = .908$ ). Our results show that training via tablet was successful in Group 1. Group 2 showed no direct effect of the training. The lack of improvement in Group 2 may be due to the complexity of the structures targeted in Relative Clauses and Wh-questions activities. For this kind of structures training activities may need more time to be assimilated, a different type of reinforcement and/or a different type of computationally mediated intervention. Further studies with more participants are needed to confirm these results.

## 4 CONCLUSION AND FUTURE WORKS

In this paper, we contribute to the design and development process of robot and computer based activities for training and assessing linguistic capabilities of children with structural language impairments (e.g., DLD and ASD). Our design was informed by a theoretically grounded psycholinguistic framework which enables us to abstract the activities into different layers Design Pattern (named as *Activity Pattern*). Our pattern-based approach promotes design reuse and extensibility at both UX and implementation levels. In addition, we validated our approach in an experimental study. Even if still preliminary, our results showed that the training via computer-based activities was effective.

The next steps of our empirical study will be to evaluate also production activities for computer-based solutions. Then, we will explore children performance on linguistic activities (comprehension and production) during interaction with robotic platforms. From a comparison analysis, we expect that the interaction with robot will boost not only language abilities but also extra-linguistic ones in children, more than computer-based interventions. In the future, exploiting our platform-agnostic feature, we will explore the

use of *Activity Pattern* also for other technological solutions, such as virtual reality (VR), augmented reality (AR), and smart spaces.

## ACKNOWLEDGMENTS

We would like to thank all the children and the families that participated in the study, the speech language therapists at the centers and our colleagues, especially Francesca Panzeri.

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